



Azure Cloud

Orchestration & Monitoring

PL_Master

runs every step in order

QualityCheck

Azure Function



Azure Data Factory

2



API
CONNECTION

4



Clean file

3

Calls



Azure Function

OpenAlex + KFUPM

Load / Raw

Data Sources

1



KAUST Repository
CSV • 2023 , API



KSU Open Data Portal
JSON • 2023–2025



KFUPM Pure
MODS XML • OAI-PMH



Crossref

Crossref REST API
JSON • 2024–2026



OpenAlex
JSON • REST API
KAU • KKU • PSAU

Data Governance



Data quality

Validation rules V1–V9

Rejected rows kept with reason

Checked before publishing

Stops if data drops 20%

Raw

Raw 1

Every

Enrich

UpdateStats

Azure Function

Logic App

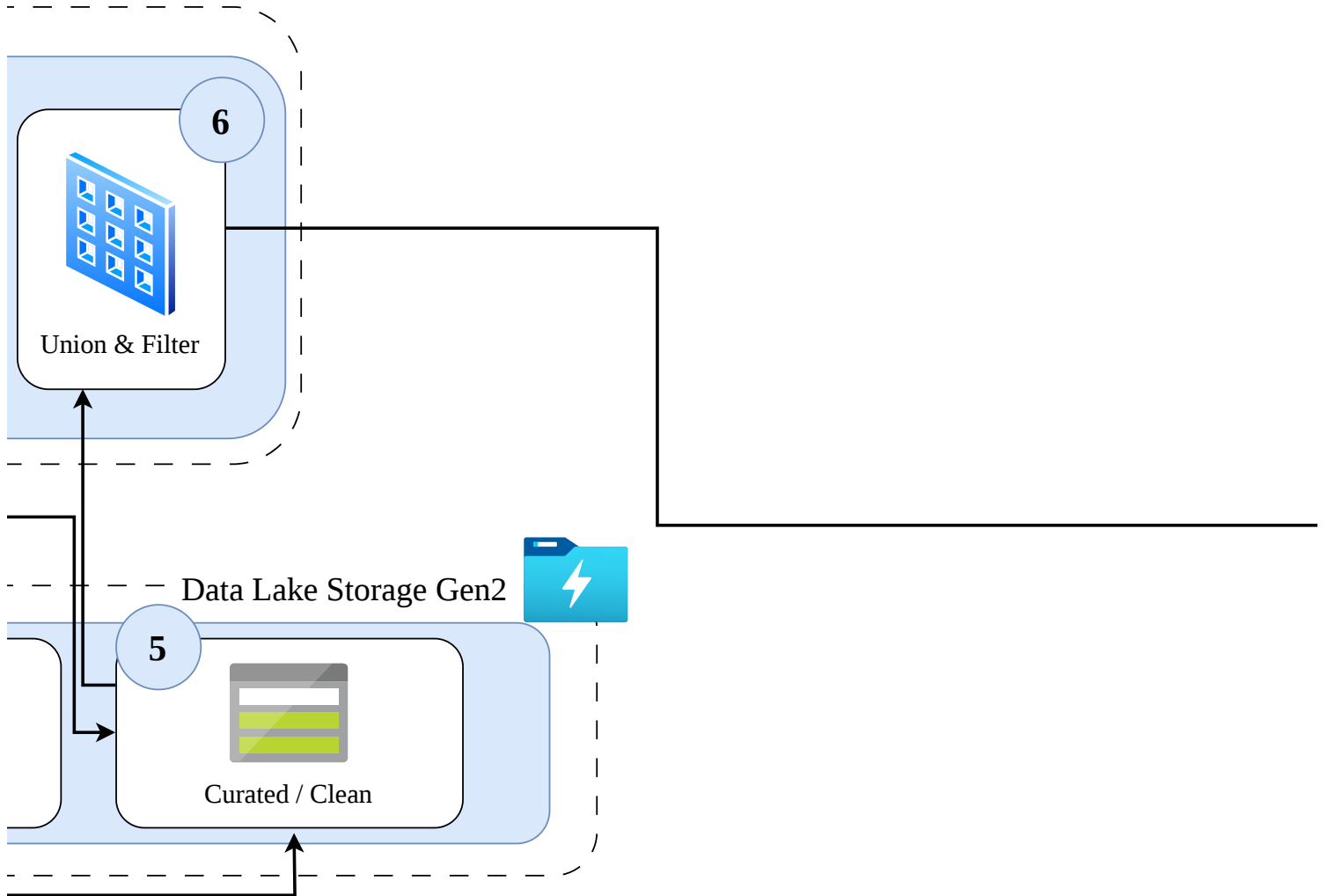
success / failure
email

Azure Monitor

failed-run alert

Key Vault

email URL se



Lineage

→ Cleaned → PostgreSQL
files never edited
y row keeps its source
chment has evidence file



Access

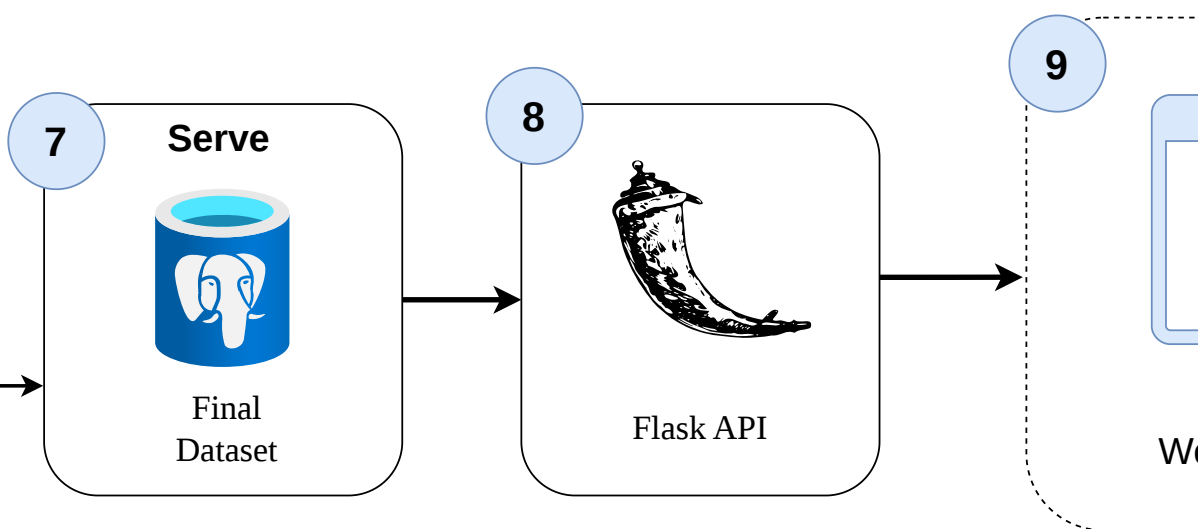
Roles: Owner, Contributor, Reader
Functions use managed identity
No stored keys
Readers: SELECT only

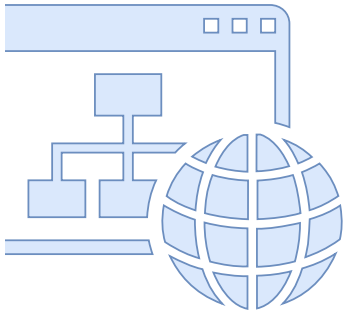


Change & audit

Pipelines versioned in GitHub
Run history in Data Factory
Quality checks logged
Stats saved per run

t
ecret





Website Dashboard





